16.1

(a) $P(AUTO=1 \mid DTIME=1) = \Lambda(-0.2376 + 0.5311 \times 1)$

$$= \Lambda(0.2935)$$

$$= \frac{1}{1+e^{-0.2935}}$$

$$= 0.5729$$

(b) $P(AUTO=1 \mid DTIME=1) = \Phi(-0.0644 + 0.3(1))$

$$= \Phi(0.2356)$$

$$= 0.5931$$

It is very close to the logit estimate of 57.29%

(c) $DTIME=3$, $\frac{dp}{dDTIME} = \lambda(l)\tilde{\gamma}_2$

$$= \frac{e^{-l}}{(1+e^{-l})^2} \cdot \tilde{\gamma}_2$$

$\because l = -0.2376 + 0.5311(3) = 1.3557$

$$\Rightarrow \frac{dp}{dDTIME} = \frac{e^{-1.3557}}{(1+e^{-1.3557})^2} \cdot 0.5311 = 0.0865$$

linear probability model: $\frac{dp}{dDTIME} = \tilde{\gamma}_2 = 0.0703$

Logit 的 ME 是 8.65%，比 LPM 的 7.03% 大. LPM marginal effect is constant, logit 的 ME 会随 DTIME 改变.

(d) $DTIME = -5$

logit: $l = -0.2376 + 0.5311(-5) = -2.8931$

$$\frac{dp}{dDTIME} = \frac{e^{2.8931}}{(1+e^{2.8931})^2} \cdot 0.5311 = 0.0264$$

Probit: $z = -0.0644 + 0.3(-5) = -1.5644$

$$\frac{dp}{dDTIME} = \phi(-1.5644) \cdot 0.3$$

$$= \frac{1}{\sqrt{2\pi}} e^{-(-1.5644)^2/2}(0.3)$$

$$= 0.0352$$

Probit ME is slightly layer than logit ME, but both are small because automobile travel time is initially much longer than bus travel time.

16.4

(a) $P(y=1 \mid x=1.5) = \Lambda(-1.836+3.021(1.5))$

$= \Lambda(2.6955)$

$= \dfrac{1}{1+e^{-2.6955}}$

$= 0.9368$

(b) $\hat{p} = 0.9368$, $\hat{p} > 0.5 \Rightarrow \hat{y}=1$

The actual outcome is $y=1$, the prediction is correct.

(c) $\gamma_1 = -1,\ \gamma_2 = 2$

$(y_1=1,\ x_1=1.5) = \Lambda(-1+2(1.5)) = \Lambda(2)$

$(y_2=1,\ x_2=0.6) = \Lambda(-1+2(0.6)) = \Lambda(0.2)$

$(y_3=0,\ x_3=0.7) = 1-\Lambda(-1+2(0.7)) = 1-\Lambda(0.4)$

$L(-1,2) = \Lambda(2)\cdot\Lambda(0.2)\cdot(1-\Lambda(0.4))$

$= \dfrac{1}{1+e^{-2}} \cdot \dfrac{1}{1+e^{-0.2}}\left(1-\dfrac{1}{1+e^{-0.4}}\right)$

$= 0.8808 \cdot 0.5498 \cdot 0.4013$

$= 0.1943$

MLE : $\ln L(-1,2) = -1.612 \Rightarrow L(-1,2) = e^{-1.612} = 0.1995$

$0.1995 > 0.1943$, MLE 的 likelihood 比 $\gamma_1=-1,\ \gamma_2=2$ 時更大

(d) True. The likelihood is the probability of observing the sample data. Since each probability term is between 0 and 1, their product is also between 0 and 1.

(e) True. $\because 0 < L < 1 \Rightarrow \ln L < 0$

16.18

(a)

```
               Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)  0.68849128  0.22850643   3.0130 0.0026525 **
lvr          0.00162385  0.00067523   2.4049 0.0163601 *
ref         -0.05932367  0.02402561  -2.4692 0.0137098 *
insur       -0.48158489  0.03036944 -15.8575 < 2.2e-16 ***
rate         0.03437614  0.00981944   3.5008 0.0004845 ***
amount       0.02376801  0.01445089   1.6447 0.1003398
credit      -0.00044190  0.00020728  -2.1319 0.0332620 *
term        -0.01261951  0.00355596  -3.5488 0.0004051 ***
arm          0.12832392  0.02769325   4.6338  4.07e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The signs are mostly reasonable. Higher LVR, higher interest rates, larger loan amounts and ARM are associated with higher delinquency risk, while higher credit scores reduce delinquency risk.

(b)

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.665303   1.992157   0.836 0.403195
lvr          0.013519   0.008199   1.649 0.099164 .
ref         -0.525064   0.230491  -2.278 0.022725 *
insur       -3.122105   0.216906 -14.394  < 2e-16 ***
rate         0.308656   0.082771   3.729 0.000192 ***
amount       0.217606   0.113769   1.913 0.055786 .
credit      -0.003564   0.001933  -1.844 0.065232 .
term        -0.133037   0.034597  -3.845 0.000120 ***
arm          1.405649   0.371355   3.785 0.000154 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 998.03  on 999  degrees of freedom
Residual deviance: 666.29  on 991  degrees of freedom
AIC: 684.29

Number of Fisher Scoring iterations: 5
```

The sign are generally the same as in the LPM, but the significance levels differ slightly.

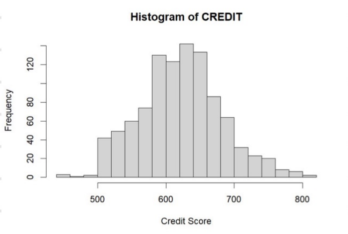
(c)

```
> pred_lpm_c
      500      1000
0.1827828 0.5785296
> pred_logit_c
      500      1000
0.1335848 0.6265535
```

The 500th observation has a relatively low predicted delinquency probability, while the 1000th observation has a much higher predicted probability.

(d)

```
result_d
credit       LPM     Logit
   500 0.5532956 0.4967030
   600 0.5091060 0.4086308
   700 0.4649165 0.3260580
```



As credit score increases, the predicted probability of delinquency decreases. The LPM decreases linearly, while the logit model decreases nonlinearly.

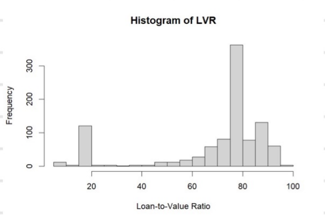
(e)

```
result_e
credit          LPM_ME      Logit_ME
   500 -0.0004418953 -0.0008910620
   600 -0.0004418953 -0.0008613439
   700 -0.0004418953 -0.0007832567
```

The LPM ME is constant, the logit ME depends on the predicted probability.

(f)

```
result_f
lvr       LPM     Logit
 20 0.4116749 0.2349087
 80 0.5091060 0.4086308
```



A higher LVR increases the predicted probability of delinquency.

(g)

```
> correct_lpm
[1] 0.858
> correct_logit
[1] 0.854
```

The two models have very similar prediction accuracy. The LPM performs slightly better, but the difference (0.4%) is very small.

(h) No. 0.5 threshold is not necessarily the best decision rule. The lender should choose the threshold based on the relative costs of approving a delinquent borrower and rejecting a good borrower. If delinquency is costly, a lower threshold such as 0.2 or 0.3 may be more appropriate.